

GROUND WATER RECHARGE

MODULE - 3

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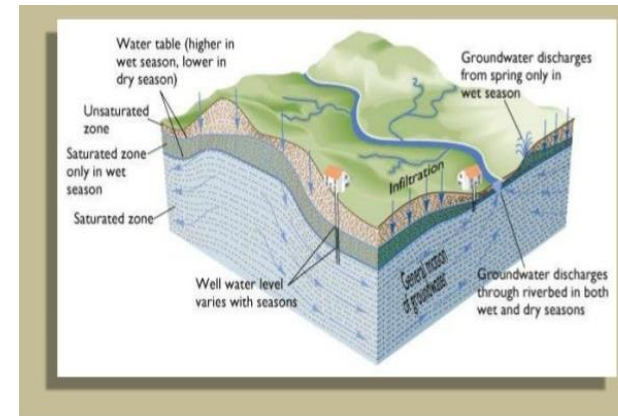
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Ground water recharge

- Groundwater recharge is the process by which water moves from the surface into underground aquifers.
- This process occurs naturally through precipitation, rivers, lakes, and artificial methods such as rainwater harvesting and recharge wells.
- Recharge is essential for maintaining groundwater levels, preventing depletion, and ensuring sustainable water supply for drinking, agriculture, and industry.

Importance of Groundwater Recharge

- Prevents groundwater depletion.
- Reduces the risk of land subsidence.
- Enhances water availability for agriculture and drinking.
- Mitigates drought effects.
- Reduces contamination by diluting pollutants in aquifers.



Types of Groundwater Recharge

1.Natural Recharge

1. Precipitation: Rainwater and melted snow infiltrate the soil.
2. River and Stream Seepage: Water from surface water bodies percolates into aquifers.
3. Floodplain Infiltration: Floodwaters seep into the ground, replenishing aquifers.

2.Artificial Recharge

1. Rainwater Harvesting: Collecting and directing rainwater into recharge pits or wells.
2. Recharge Wells: Directly injecting water into aquifers.
3. Percolation Ponds: Constructed basins that allow water to seep into the ground.

Factors affecting ground water recharge

1. Climatic Factors

- **Precipitation** – More rainfall increases recharge, while arid regions experience minimal recharge.
- **Temperature** – High temperatures increase evaporation, reducing the amount of water available for infiltration.
- **Humidity** – Higher humidity reduces evaporation, aiding recharge.
- **Wind Speed** – Strong winds increase evaporation and reduce infiltration.

2. Soil and Geology

- **Soil Permeability** – Sandy and loamy soils allow better infiltration, while clayey soils restrict water movement.
- **Rock Type** – Porous rocks like limestone and sandstone enhance recharge, while impermeable rocks like granite hinder it.
- **Depth to Water Table** – A deep water table requires more infiltration before recharge occurs.

3. Land Use and Vegetation

- **Vegetation Cover & Type** – Plants slow runoff, allowing more water to seep into the ground. Deforestation reduces recharge.
 - Legumes crops with more recharge
 - Rice, cotton, sugarcane etc., less recharge
- **Urbanization** – Paved surfaces prevent infiltration, leading to more runoff and less recharge.
- **Agricultural Practices** – Overuse of groundwater and poor irrigation practices can reduce recharge, while conservation techniques like contour farming improve it.



4. Surface Water Bodies

- **Lakes and Rivers** – Can provide recharge through seepage if the water table is lower than the water body.
- **Artificial Reservoirs and Ponds** – Help retain water for gradual infiltration.

5. Human Activities

- **Groundwater Extraction** – Excessive pumping lowers the water table, reducing natural recharge rates.
- **Recharge Structures** – Artificial recharge methods like check dams, percolation ponds, and recharge wells enhance groundwater levels.
- **Pollution and Contamination** – Contaminants can block soil pores, reducing infiltration.

Traditional techniques of water harvesting

Many ancient methods have been effective for centuries, especially in arid and semi-arid regions



Tanka:

A tanka (*tank*) is a small underground or surface-level tank used to collect and store rainwater, often from rooftops or paved catchments

- Circular or rectangular underground tank, plastered with lime, cement, or stone to prevent seepage.
- Catchment area - roof or a sloped surface (plastering a mixture of clay, silt of pond, coal ash and gravel)
- Capacity typically ranges from 2,000 to 25,000 liters, depending on the household's need and catchment area.
- The tank is covered to prevent evaporation and contamination.
- Found in **Rajasthan**.
- These structures are used for collecting rainwater which is used by human and livestock for drinking.



Khadin:

Khadin is a technique which conserves water and soil, which is designed to harvest surface runoff water for agriculture particularly in very low fall areas

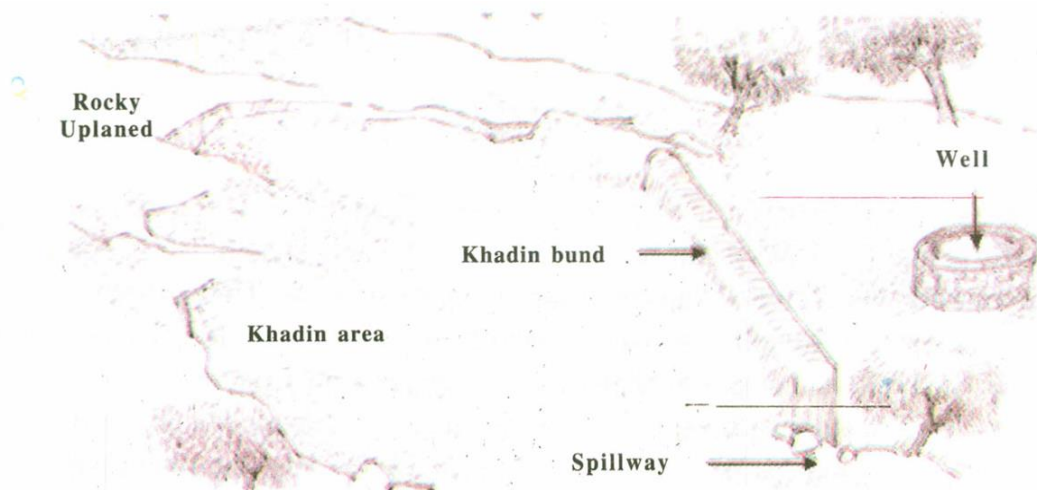
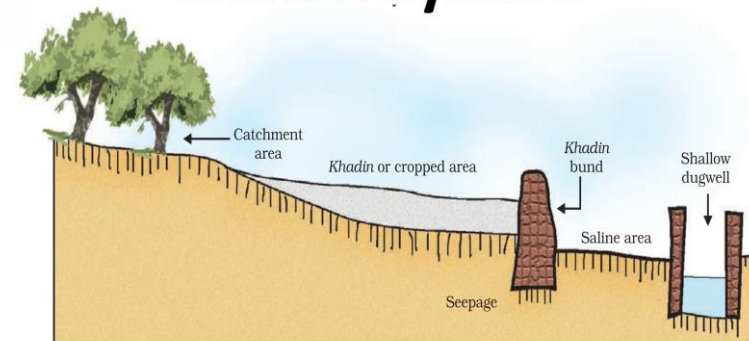


Fig.1.7: Schematic diagram of a Khadin



Khadin System



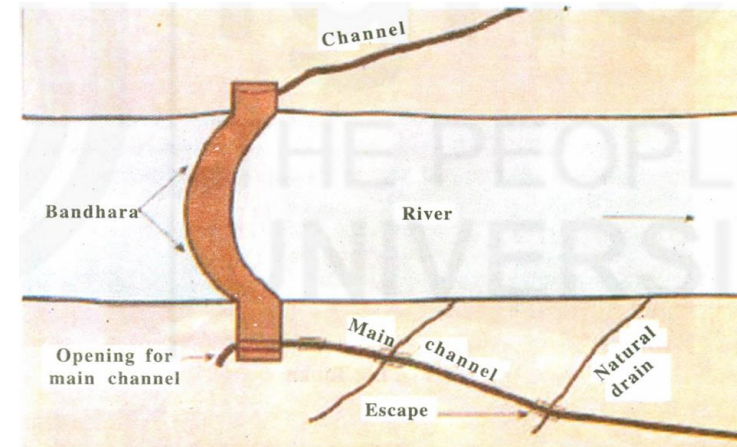
- It consists of an earthen embankment or bund (100- 300 m) built across the lower hill slopes lying below gravelly uplands.
- You can also observe that sluices and spillways are provided to drain off excess water.
- The khadin system is based on the principle of harvesting rainwater on farmland and subsequent use of this water for providing life saving irrigation to crops.

Advantages:

- Makes use of rainfall in dry regions for agriculture.
- Increases groundwater recharge.
- Prevents soil erosion and improves soil fertility.
- Ideal for sustainable farming in deserts.

Bandharas:

- Bandharas is another traditional system of water harvesting for irrigating crops especially sugarcane in Maharashtra
- A bandhara is a low, earthen, masonry, or concrete dam constructed across a river or stream.
- It helps slow down the water flow, allowing it to percolate into the ground while creating a small reservoir upstream.
- These structures are also very useful to control floods during heavy rain.
- A system weir vents has been developed: The shutters are provided which are kept open during heavy rainfall.
- Surplus water passes through gates which help in avoiding floods.



Ahars:

- These water harvesting structures are generally found in East U.P. and Bihar
- This system based on slope is similar to khadin.
- The bunds are constructed with 1:2 ratio of upstream and downstream slope.
- The earthen bunds are about 3 m high, Runoff water is collected behind the bund.
- A spillway is provided , to release surplus water and avoid damage to earthen bund.
- Concrete or cast iron pipes with diameter of 15-30 cm are fitted in bunds at an interval of 60-80 meter distance for irrigation purposes.
- Gates are also provided to drain out water from the bund in sufficient quantity for pre sowing irrigation.



Pat system:

- The **Pat System** (also spelled *Patt system*) is a traditional water harvesting and irrigation method found in parts of **central India**, especially in tribal and hilly regions like **Madhya Pradesh** and **Chhattisgarh**.
- The Pat system uses the natural slope of the land to divert water from hill streams or rivers to agricultural fields.
- It operates without pumps or modern machinery—just the force of gravity.
- Water is diverted using small temporary bunds (dams) made of stones, earth, and vegetation.



Zing system:

- Zings, found in Ladakh, are small tanks that collect melting glacier water
- A network of guiding channels brings water from the glacier to the tank..
- A trickle in the morning, the melting waters of the glacier turn into a flowing stream by the afternoon.
- The water, collected by evening, is used in the fields on the following day.
- Made using stones and clay, sometimes with a lining to reduce seepage.



Baois system:

- Traditional and architecturally significant water harvesting system found primarily in northern and western India, especially in Rajasthan, Gujarat, Delhi, and Madhya Pradesh.
- A deep, multi-storeyed well with steps leading down to the water.
- It serves as both a water storage structure and a cool resting place in hot, arid climates.
- Deep well with descending steps and platforms, often beautifully carved.
- Built with stone, bricks, lime plaster.
- Steps allow access to water even as the level fluctuates



Calculation of available rain for harvesting:

The amount of runoff generated from the total amount of rainwater received over an area depends upon:

- The nature of the catchment or rooftop,
- Intensity of rainfall and
- Collection efficiency of the system without significant losses.

The maximum amount of water that can be collected and harvested from an area is called the Water Harvesting Potential.

$$\text{Water Harvesting Potentials} = (\text{Rainfall}) \times (\text{Collection Efficiency}) \times (\text{Rooftop or catchment area})$$

Example:

Determine the volume of water harvested for a building of roof area of 100 m^2 located in an area receiving rainfall of about 600 mm per annum. Assume collection efficiency of 70% (include runoff coefficient and spillage and leakages from pipes).

Solution:

Rooftop area = 100 m^2

Rainfall depth = 600 mm

Volume of water harvested = $0.60 \times 100 \times 0.70$

= 42 m^3

= **42000 litres** (i.e. $1 \text{ m}^3 = 1000 \text{ litres}$)

Size of Storage Tanks for Rural Areas:

Size of the storage tank should be carefully selected considering various factors such as:

- Number of persons in the household,
- Water use,
- Duration of water scarcity,
- Rainfall,
- Type and size of house roof
- The status of existing water sources in the area.

In general, the period of water scarcity for domestic purposes ranges between 90 and 200 days depending upon the quantity and distribution of rainfall and water sources existing in the area.

The water use of the household needs to be estimated " based on the local culture and habits of the people influencing the water use, Minimum, quantity of per capita water needed for survival should be around 50 litre per day.

The size of water storage tank may be determined using the following relation.

Size of storage tank based on water requirement (litre) = (Number of persons in the household) x (Period of water scarcity (in days)) x (Per capita water requirement (litre per day))

The capacity of storage tank, which reflects the total household water requirement during the period of water scarcity should be compared with the amount of water available from house rooftop during rains’.

If the amount of water available from rooftop rainwater harvesting is less than the required capacity of storage tank, then the household will have to use the water available from roof only for a part of the water scarcity period.

Water available from roof is obtained from the following relation:

$$\text{Water available (litre)} = \text{Annual rainfall} \times \text{Roof area} \times \text{Runoff coefficient}$$

Runoff coefficient depend on the type of roof surface and indicates the fraction of rainwater that can be collected from a particular roof. Runoff coefficients for common types of roofs are shown in Table.

Surface type	Runoff coefficient
GI Sheet	0.90
Asbestos	0.80
Tiled	0.75
Concrete	0.70

Example:

Determine the size of storage tank for the following data:

No. of persons in the selected household (4adults and 4 children) = 8

Period of water scarcity for the domestic needs = 120 days

Per capita water requirement = 50 litre/day

Annual average rainfall = 1000mm

Area of roof made of country tiles = 20 m²

Runoff coefficient for tiled roof = 0.75

Solution:

Size of storage tank (litre) water = Number of persons in a household x Period of scarcity (days) x Per capita water requirement (litre/day)

$$= 8 \times 120 \times 50 = 48000 \text{ litre}$$

Water availability from rooftop = Annual rainfall x Area of roof x Coefficient of runoff for the roof

$$= 1 \times 20 \times 0.75 = 15 \text{ m}^3 = 15000 \text{ litre}$$

Thus the harvested water meets only part of the water requirement:

$$= (15000/48000) \times 100 = 31 \%$$

Size of Storage Tank for Urban Area:

- In major cities and towns where regular water supply already exists throughout the year, the size of the storage tank for harvesting roof top rainwater can be decided as per the quantum of rainfall occurring in a single event.
- It is assumed that serious water scarcity period is not considered and collection of rooftop rainwater is voluntary/ mandatory as an effort towards saving our natural resources.
- Moreover, it is difficult to create enough storage space due to various constraints.
- In such a situation, it is ideal to create the storage space to collect the rainfall for an event and utilize the same before the next spell